Graphs 7 — Using graphs to solve equations · Non-linear simultaneous equations

Calculator allowed; graph paper needed. Marks in brackets. ★ = harder question. Rearrange the equation so one side is the curve you have drawn; the other side is the line to draw.

1 **a** Copy and complete the table and draw the graph of $y = x^2 - 3x$ for $-1 \le x \le 4$. [8]
b By drawing suitable straight lines on your graph, solve **i** $x^2 - 3x = 2 - x$ **ii** $x^2 - 4x + 1 = 0$.

x	-1	0	1	2	3	4
y	4		-2			4

2 The graph of $y = 2x^2 - 5x + 1$ has been drawn. Find the equation of the straight line that should be drawn to solve each equation. [6]
a $2x^2 - 5x - 3 = 0$ **b** $2x^2 - 7x + 1 = 0$ **c** $2x^2 - 3x - 4 = 0$

3 The graph of $y = x^3 + 2x^2 - 3x - 1$ has been drawn. Find the equation of the line to draw to solve [4]
a $x^3 + 2x^2 - 3x + 4 = 0$ **b** $x^3 + 2x^2 - 5x - 3 = 0$

4 **a** Copy and complete the table and draw the graph of $y = 6 + x - x^2$ for $-2 \le x \le 4$. [7]
b Use your graph to solve the simultaneous equations $y = 6 + x - x^2$ and $x + 2y = 6$, giving your answers to 1 d.p.

x	-2	-1	0	1	2	3	4
y	0		6		4		-6

Total: 25 marks

Check: the drawn curve must appear on one side exactly as written — add/subtract the same thing to both sides · the solutions are the x-coordinates where line and curve cross · two crossings → two solutions · for simultaneous equations read both coordinates of each crossing.